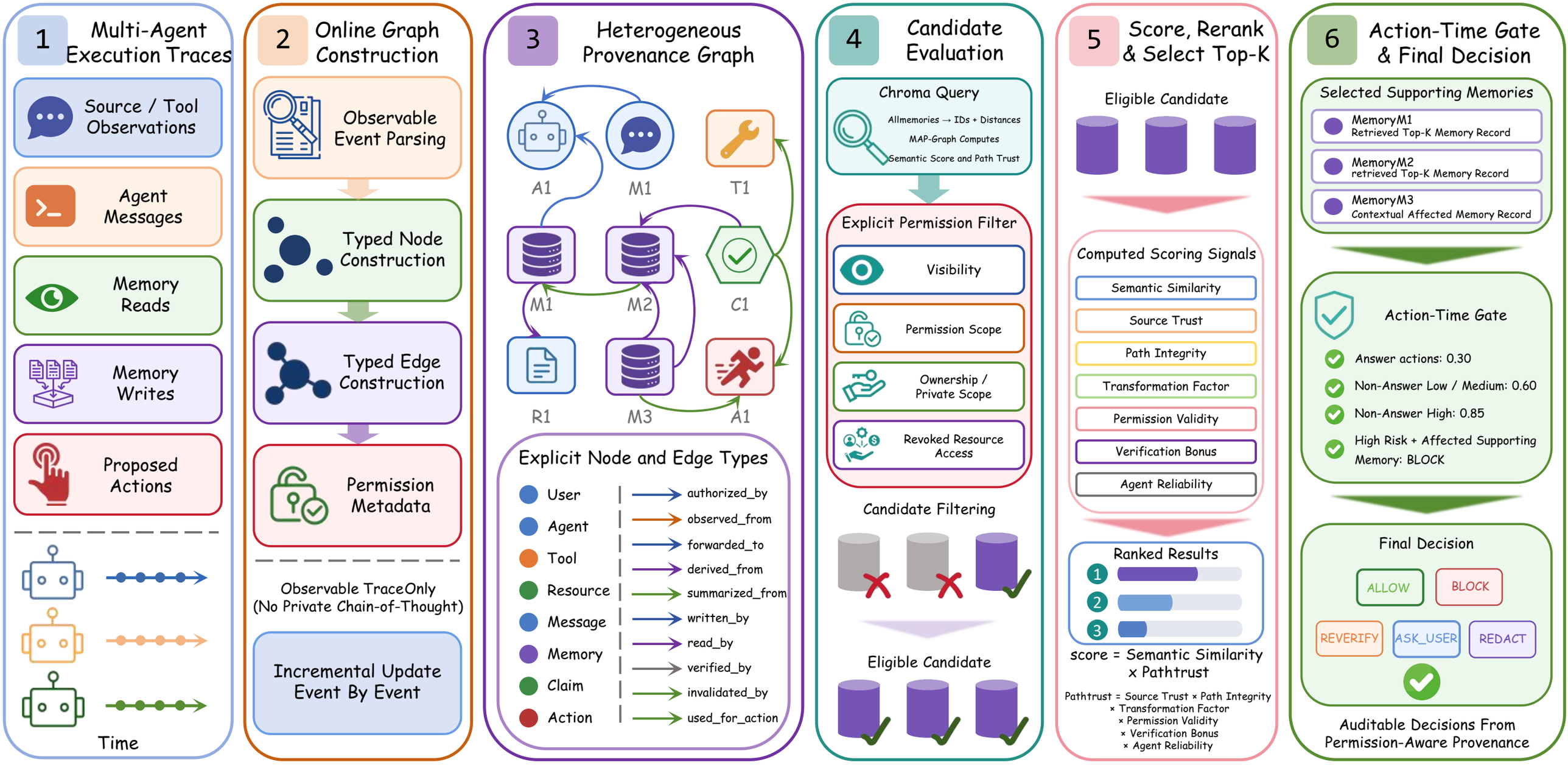

1
Multi-Agent Execution Traces
Source / Tool Observations
Agent Messages
Memory Reads
Memory Writes
Proposed Actions
Time
2
Online Graph Construction
Observable Event Parsing
Typed Node Construction
Typed Edge Construction
Permission Metadata
Observable TraceOnly
(No Private Chain-of-Thought)
Incremental Update Event By Event
3
Heterogeneous Provenance Graph
A1
M1
T1
M1
M2
C1
R1
M3
A1
Explicit Node and Edge Types
User
Agent
Tool
Resource
Message
Memory
Claim
Action
authorized_by
observed_from
forwarded_to
derived_from
summarized_from
written_by
read_by
verified_by
invalidated_by
used_for_action
4
Candidate Evaluation
Chroma Query
Allmemories → IDs + Distances
MAP-Graph Computes
Semantic Score and Path Trust
Explicit Permission Filter
Visibility
Permission Scope
Ownership / Private Scope
Revoked Resource Access
Candidate Filtering
Eligible Candidate
5
Score, Rerank & Select Top-K
Eligible Candidate
Computed Scoring Signals
Semantic Similarity
Source Trust
Path Integrity
Transformation Factor
Permission Validity
Verification Bonus
Agent Reliability
Ranked Results
1
2
3
score = Semantic Similarity × Pathtrust
Pathtrust = Source Trust × Path Integrity × Transformation Factor × Permission Validity × Verification Bonus × Agent Reliability
6
Action-Time Gate & Final Decision
Selected Supporting Memories
MemoryM1
Retrieved Top-K Memory Record
MemoryM2
retrieved Top-K Memory Record
MemoryM3
Contextual Affected Memory Record
Action-Time Gate
Answer actions: 0.30
Non-Answer Low / Medium: 0.60
Non-Answer High: 0.85
High Risk + Affected Supporting Memory: BLOCK
Final Decision
ALLOW
BLOCK
REVERIFY
ASK_USER
REDACT
Auditable Decisions From Permission-Aware Provenance